World Journal of Surgical Oncology

RESEARCH

Open Access

Analysis of bone metastasis multidisciplinary team meeting activity in a university hospital: patient profiles, prognostic factors

Paul Chanzy[1,2], Marc Khalifé[3,4], Laetitia Morardet[5], Hugues Pascal-Moussellard[1,2], Raphael Bonaccorsi[1,2] and Laura Marie-Hardy[1,2*]

Abstract

Background Bone metastases a represent a challenge in the oncological management of patients as the complications can be severe, as fractures or neurological impairment in spinal metastases. Recommendations on the management of bone metastases are constantly being improved. However, there are few recent prognostic data on bone metastases in the literature to assess the impact of advances in oncological treatment on patient survival.

Hypothesis An analysis of the activity of an MDT (Multidisciplinary Team Meeting) over 2 years could provide recent data regarding prognosis of patients presenting with bone metastasis.

Materials and methods This is a retrospective study of patients presented at an MDT from 2020 to 2021 in a university hospital. The objectives were to describe a representative cohort of patients with bone metastasis and the therapeutic decisions made (cementoplasty/radiotherapy/surgery) and patient survival according to tumor type, location of bone metastasis (spine/pelvis/other).

Results Nine hundred ninety-two presentations of 630 patients were analyzed. The most frequent primary cancer was bronchopulmonary, followed by breast, myeloma, prostate and thyroid. 85% of patients had a spinal lesion, 51% a pelvic lesion and 52% another bone site. 49% of patients underwent radiotherapy, 39% cementoplasty and 22% surgery. Median survival was 17 months. Bronchopulmonary and prostate tumors patients had a poorer prognosis. 37% of operated patients died within 6 months of follow-up.

Conclusion This study provides a description of bone MDT meeting activity in a university hospital setting, and gives a picture of the volume of activity, prognostic factors and decisions taken with significant recourse to cementoplasty, compared with older series. The presence of bone metastases still represents a turning point in the progression of oncological diseases.

Level of evidence IV, retrospective study.

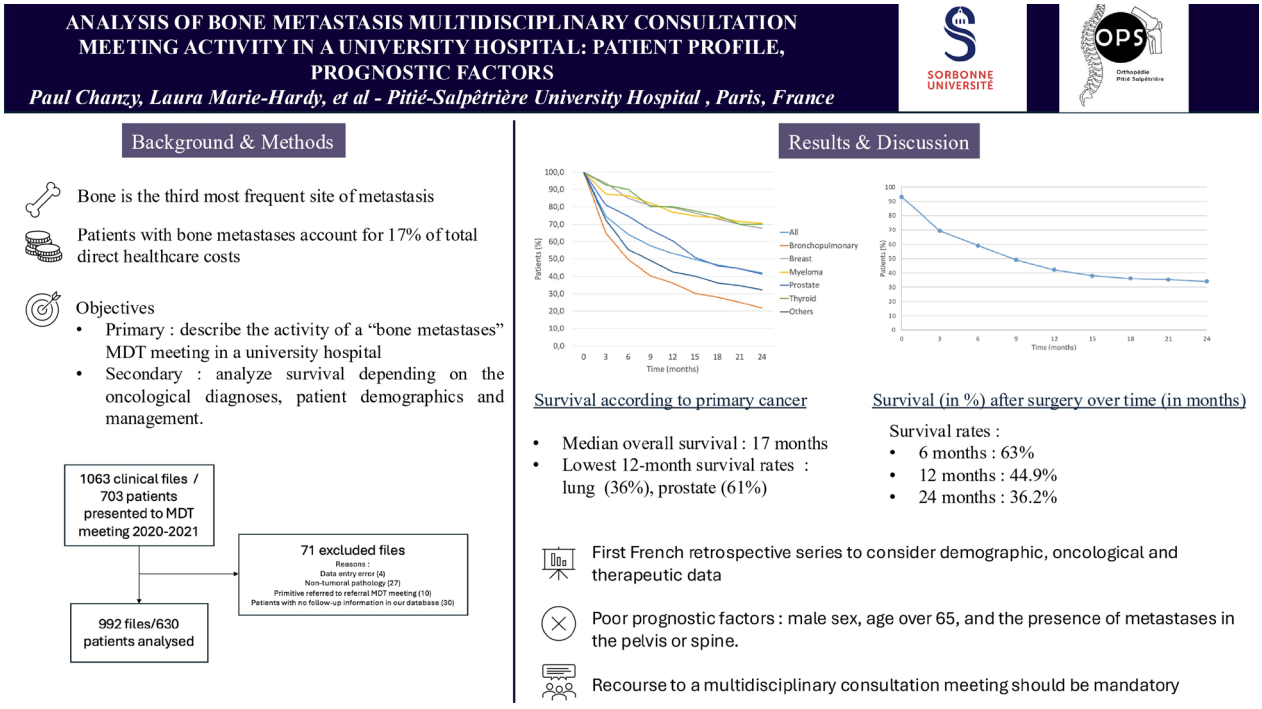
Keywords Follow-up study, Bone neoplasms, Cementoplasty, Prognosis study, Metastasis

*Correspondence:
Laura Marie-Hardy
l.mariehardy@gmail.com

Full list of author information is available at the end of the article

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Graphical Abstract



Introduction

In France, nearly 400,000 new cancer cases are diagnosed annually [1]; of these, 60% will develop metastases, including 10% with spinal involvement [2–4]. Bone is the third most common site of metastasis, after the liver and lungs [5], and bone metastases are the leading cause of cancer-related pain [6]. These metastases pose significant challenges in patient management due to the risk of neurological complications (in cases of spinal involvement), fracture, and the resulting painful functional impairment. Metastatic bone disease also represents a substantial burden on society, the economy, caregivers, and healthcare systems [7].

The primary challenge lies in tailoring therapy to the patient's prognosis. However, accurate prognostic prediction remains difficult in clinical practice, despite the development of increasingly precise composite oncology scores. Estimated survival is often overestimated by several months [8–11]. While 60% of bone metastases respond well to radiotherapy [12], clinical improvements—particularly in pain relief—may take several weeks to manifest [13–15]. Reported post-radiotherapy survival medians vary widely, ranging from a few weeks to several months [16–18], and depend mainly on the patient's general condition (WHO performance status), the presence of visceral metastases, adjuvant chemotherapy [19], and biological parameters such as C-reactive protein, albumin, and blood calcium levels, as highlighted by Prost et al. [20]. Percutaneous cementoplasty offers an effective treatment option for lytic, painful lesions that are at risk of—or have already—fractured, especially in the spine, due to its stabilizing properties and minimally invasive nature [21]. Although surgery can effectively stabilize lesions, it carries inherent risks, which are heightened in this fragile patient population.

The complexity of managing these patients necessitates a multidisciplinary team (MDT) approach to ensure optimal support and treatment [22]. MDT meetings involve at least three medical specialties and facilitate collegial decision-making regarding patient management. These meetings may be general oncology discussions or specialized, such as the "bone metastases" MDT.

Patient survival in the context of bone metastases is highly heterogeneous, primarily driven by tumor type and genetic mutation profiles. While the prognosis for certain cancers has improved significantly in recent years—largely due to the advent of targeted therapies [23]—clinical guidelines for the management of bone metastases continue to evolve rapidly [24]. However, the literature lacks recent, comprehensive prognostic data on skeletal-related adverse events (SREs) in this population. This gap impedes clinicians' ability to access evidence-based guidance and assess how advances in oncological treatments influence patient survival, both of which are critical for informed decision-making.

The primary objective of this study was to describe the activity of a multidisciplinary team (MDT) meeting focused on bone metastases within a university hospital setting. The secondary objective was to analyze survival outcomes based on the most frequent oncological diagnoses associated with bone metastases, as well as patient demographics and management strategies.

Materials and methods

Study design

This was a retrospective, single-center, observational study registered in the institutional registry under number 20,240,410,163,136. All patients presented at the "bone metastases" multidisciplinary team (MDT) meeting of a French university hospital between January 1, 2020, and December 31, 2021, with secondary bone lesions, were included retrospectively.

The bone metastases MDT is a dedicated weekly meeting focused on bone lesions. Its purpose is to determine whether the presented bone metastases require additional local treatment (such as surgery, cementoplasty, or radiotherapy) or can be managed with systemic oncologic therapy alone (which would then be discussed in a specific oncology MDT). The referring oncologist or medical specialist (e.g., gynecologist for breast tumors) completes a standardized form with relevant clinical information, and imaging is accessible via the internal hospital system. In our institution, it is standard practice for all patients with significant bone metastases to be presented to the MDT.

Bone metastases are considered non-significant only if they meet all of the following criteria:

- Extra-spinal location,
- No fracture risk,
- Non-painful,
- Presence of other known prior bone metastases.

The "bone metastases" MDT meeting includes at least three specialties involved in the management of patients with secondary bone lesions: orthopedic surgeons, oncologists, radiation therapists, and interventional radiologists.

Patients presented at the MDT meeting with non-tumoral pathology (e.g., osteoporosis) or primary malignant bone lesions (e.g., osteosarcoma) were excluded from the analysis. Incomplete clinical records were also excluded, as were patients with no follow-up information in the database. When a patient was presented multiple times at the MDT meeting, only the data from the first presentation were retained for analysis. If a patient was presented for multiple threatening lesions (e.g., spine and pelvis), all relevant lesions were recorded and considered in the analysis. The minimum follow-up duration was 24 months, and the maximum was 43.8 months.

Analysis criteria

Clinical records were reviewed one by one by two orthopedic surgeons, in order to collect the following data:

- Demographic data: date of birth, sex, age at presentation to MDT meeting.
- Oncological data: histological type of the primary tumor (bronchopulmonary, breast, myeloma, prostate, thyroid, other), site of bone metastases (divided into spine, pelvis and limbs).
- Proposed treatment following MDT meeting:
  - Osteoarticular surgery: yes/no.
  - Radiotherapy: yes/no.
  - Interventional radiology at the site of the bone lesion: yes/no.
- Follow-up: For each patient were collected data regarding:
  - The date of death, verified for each patient via the official national database (INSEE (Institut national de la statistique et des études économiques) (https://deces.matchid.io/search).
  - And/or date of the last medical event on the patient software's medical record (consultation, test result).
  - Management of lost to follow-up patients : patients who were not recorded as deceased in the INSEE database were assumed to be alive at the time of data entry.

The MDT decisions were made by a panel of specialists, including oncologists, orthopedic surgeons, radiotherapists, and interventional radiologists. When available, the treating specialist (e.g., a hematologist for myeloma patients) presented the case at the MDT. If not, all relevant information was provided on a dedicated form. In all cases, the MDT specialists had full access to the patient's medical file.

The MDT was solely responsible for discussing the treatment of bone metastases, while the general management of the oncologic disease was determined in dedicated oncology meetings. Decisions were made at the discretion of the MDT specialists, based on the following considerations:

- Patient-related factors: Frailty, age, performance status (PS), comorbidities, and overall cancer prognosis.

- Lesion-related factors: Risk of fracture, instability, risk of neurological impairment (for spinal lesions), and pain due to the lesion.
- Oncologic data: Presence of isolated metastasis or diffuse lesions, overall disease progression or control with oncologic therapies, and type of cancer.

The analysis presented here is a retrospective review of the decisions made during the MDT meetings.

Data analysis

An initial descriptive analysis of the cohort was performed. The primary endpoint was overall survival (OS), measured in months from the date of first presentation at the multidisciplinary team (MDT) meeting (for patients presented multiple times, only the first presentation was considered). Survival was calculated from MDT presentation rather than cancer diagnosis to specifically evaluate the impact of MDT discussions on patient outcomes. Survival analyses were stratified by primary cancer type, presence of bone metastases in the pelvis or other skeletal sites, treatment modality, sex, and age.

To compare survival across groups, analyses focused on the five most common primary cancers (breast, lung, prostate, myeloma, and thyroid) and a miscellaneous category for less frequent cancers. Age was dichotomized at the clinically relevant threshold of ≤ 65 and > 65 years to facilitate the interpretation of hazard ratios (HRs). Log-rank tests were used to assess significant differences in survival between groups.

Two Cox proportional hazards models were developed. The first model estimated HRs for age, sex, primary cancer type, and the presence of metastatic lesions in the spine, pelvis, or other bones. The second model incorporated management strategies (surgery, cement augmentation, and chemoradiotherapy) to evaluate their association with survival. Variables included in the models were selected based on their association with survival in univariate analysis and their clinical relevance.

Kaplan-Meier-type [25] survival curves were obtained. Statistical analysis was performed using R software (Version 2024.09.1 + 394), with p-values considered significant if < 0.05. The interquartile range (IQR) was defined as the difference between the third quartile (Q3) and the first quartile (Q1), representing the spread of the middle 50% of the data.

Results

Participants

1063 clinical files corresponding to 703 patients were presented to the "bone metastases" MDT meeting. 71 patients were excluded for the following reasons: 4 data entry errors, 27 non-tumoral lesions, 10 primary lesions (referred to a dedicated MDT meeting) and 33 patients without follow-up information in our database (Fig. 1). A total of 992 files from 630 patients were studied. Altogether here are therefore presented the data corresponding to 630 presentations in the MDT meeting. The number of patients lost to follow-up at 1 year and 2 years was 29 and 48, respectively.

The characteristics of the study population are presented in Table 1. The Shapiro–Wilk test for age was significant ($p < 0.001$). The mean age was 65 ± 14 years (range: 19–96), and the median age was 67 years (IQR: 57–74). Overall, 24% of patients were older than 75 years, and 55.9% were men. Primary sites were divided into 6 groups: bronchopulmonary (22.1%), breast (14.8%), myeloma (14.9%), prostate (9.8%), thyroid (6.2%), and others (32.2%). Metastases to the spine (85.1%) were more frequent than to the pelvis (51.6%) or other bones (52.1%). Radiotherapy was the most frequently proposed treatment (48.7%), followed by cementoplasty (39.2%) and surgery (21.8%). A combination of therapies was used in 33,3% of patients: 6,5%

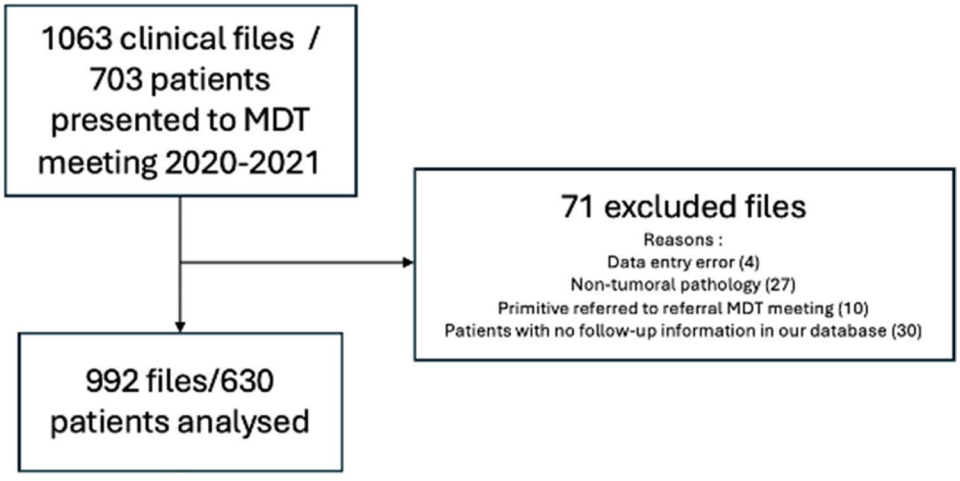


Fig. 1 Flow chart

Table 1 Baseline characteristics of the cohort (age, gender, cancer type, metastasis status and treatments); n= number of patients, SD= standard deviation

Age (mean, ± SD; in years)	65 ± 14
Gender: Male (%)/Female (%)	352 (55.9) / 278 (44.1)
Cancer type: n (%)	
Bronchopulmonary	139 (22.1)
Breast	93 (14.8)
Myeloma	94 (14.9)
Prostate	62 (9.8)
Thyroid	39 (6.2)
Others	203 (32.2)
Metastasis: n (%)	
Spine	536 (85.1)
Pelvis	325 (51.6)
Other bones	328 (52.1)
Treatments: n (%)	
Surgery	137 (21.8)
Radiotherapy	307 (48.7)
Cementoplasty	247 (39.2)
Combination of treatments: n (%)	
Surgery + radiotherapy	67 (10,6)
Surgery + cementoplasty	15 (2,4)
Radiotherapy + cementoplasty	87 (13,8)
Surgery + cementoplasty + radiotherapy	41 (6,5)

surgery-cementoplasty-radiotherapy, 10,6% surgery-radiotherapy, 2.4% surgery-cementoplasty, 13,8% radiotherapy-cementoplasty. 78,8% of patients who underwent surgery received radiotherapy after their procedure. The bone metastasis of 190 patients (30.2%) did not require specific management (either surgery, cementoplasty or radiotherapy) and patients were referred to oncologic specialist for further management and surveillance.

Survival by characteristic

Demographics

The median overall survival of the cohort was 17 months. At 24 months, survival rate was 44.3%. The survival over time is displayed as Fig. 2.

Univariate analysis highlighted male gender as a significant adverse survival criterion (35.8% vs. 55% survival rate at 24 months, $p < 0.0001$). This difference was confirmed in multivariable analysis (HR = 1.36, $p = 0.02$). Age over 65 years was significantly associated with poorer survival in univariate analysis (50.4% vs. 39.7% survival rate at 24 months, $p = 0.009$) (Tables 2 and 3). Older age was also found to be statistically associated with poorer survival in multivariable analysis (HR = 1.32, $p < 0.001$), showing that each additional year of age was associated with a 1.6% increase in the risk of death.

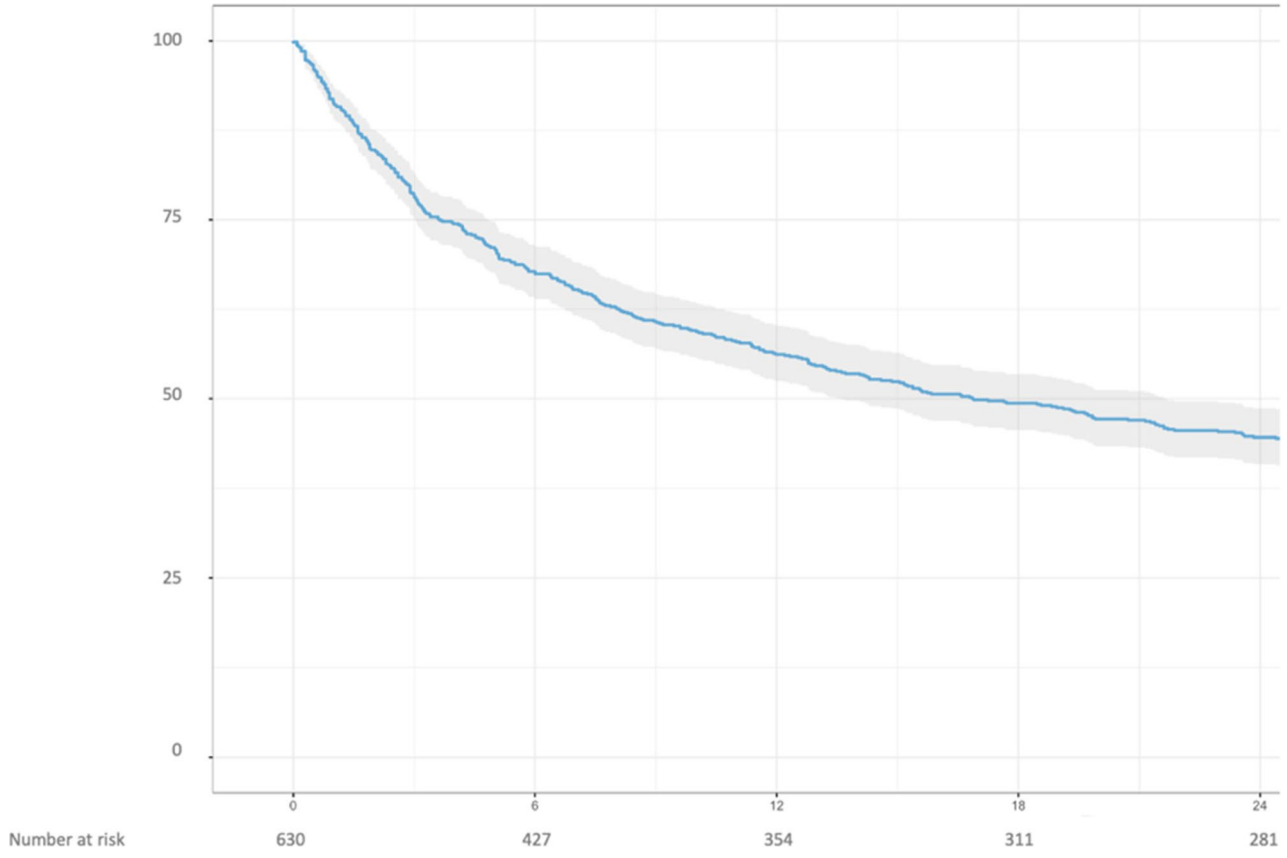


Fig. 2 Kaplan–Meier estimates of cohort survival over time (in months), % of patients alive at the analysis endpoint

Table 2 Survival according to gender, age, cancer type. Median survival (months), and 24-month survival rates are reported for each subgroup. Univariate comparisons were performed using log-rank tests. Hazard ratios were obtained though multivariable Cox models. Significant p-values (< 0.05) are denoted by an asterisk

		Median survival	Survival at 24 months	*p*
Gender	**Male**	10.6	35.8%	*0.007**
	Female	> 24	55%	
	Hazard ratio		1.36	*0.02**
Age	**< 65 years old**	> 24	50.4%	*0.2*
	≥ 65 years old	12.9	39.7%	
	Hazard ratio		1.32	*< 0.001**
Cancer type	**Lung**	5.1	21.6%	*< 0.001**
	Prostate	16.3	41.9%	
	Breast	> 24	67.7%	
	Thyroid	> 24	71.8%	
	Myeloma	> 24	71.3%	
	Other	8,4	32.2%	

Table 3 Survival according to metastasis location (spine, pelvis, or other bones) and type of management (cementoplasty, radiotherapy, or surgery). Median survival (months), and 24-month survival rates are reported for each subgroup. Univariate comparisons were performed using log-rank tests. Hazard ratios were obtained though multivariable Cox models. Significant *p*-values (< 0.05) are denoted by an asterisk

		Median survival	Survival at 24 months	*p*
Metastases	**Spinal metastases**	14.4	42.5%	*0.1*
	No spinal metastases	> 24	53.2%	
	Hazard ratio		1.53	*0.007**
	Pelvic metastases	11.9	37.7%	*0.3*
	No pelvic metastases	> 24	50.8%	
	Hazard ratio		1.57	*< 0.001**
	Other bone metastases	13.1	39.8%	*0.07*
	No other bone metastases	21.5	48.3%	
	Hazard ratio		1.38	*p* = 0.005*
Management	**Cementoplasty**	10.6	37.4%	*< 0.001**
	No cementoplasty	21.3	47.0%	
	Hazard ratio		1.25	*0.03**
	Radiotherapy	9.6	34.7%	*0.1*
	No radiotherapy	> 24	52.9%	
	Hazard ratio		1.38	*0.003**
	Surgery	10.3	36.2%	*0.4*
	No surgery	19.6	46.3%	
	Hazard ratio		1.12	*0.22*

Types of cancer and lesions

Survival outcomes differed significantly according to the primary tumor type in patients with bone metastases. Lung cancer demonstrated the poorest prognosis, with 12-month and 24-month overall survival (OS) rates of 36.0% and 21.6%, respectively. Prostate cancer also showed relatively low survival rates (61.3% at 12 months and 41.9% at 24 months). In contrast, patients with bone metastases originating from breast, thyroid, or myeloma tumors exhibited a more favorable prognosis, with 12-month OS rates ranging from 76.8% to 80.0% and 24-month OS rates between 68.0% and 72.0% (Table 2). The "other" category of primary tumors had intermediate survival rates, with 42.6% OS at 12 months and 32.2% at 24 months. Kaplan-Meier survival curves, stratified by primary tumor type, are illustrated in Fig. 3.

In the Cox model, spinal metastases were associated with poorer survival (HR = 1.53, $p = 0.007$), as were pelvic metastases (HR = 1.57, $p < 0.001$) and metastases to other bones (HR = 1.38, $p = 0.005$). (Table 2).

Survival according to management

In multivariable analysis, survival was significantly higher in the group without cementoplasty than in the group with cementoplasty (HR = 1.25 [1.02–1.53] $p = 0.03$), reflecting that radiotherapy was more frequently used in patients with a poorer prognosis. In multivariable analysis, radiotherapy was associated with shorter survival (HR = 1.38, $p = 0.003$), reflecting that radiotherapy was more frequently used in patients with a poorer prognosis. Conversely, there was no difference in survival likelihood according to surgery (HR = 1.12 [0.91–1.49], $p = 0.22$). Management with cementoplasty and radiotherapy appeared to be associated with a poorer prognosis of survival compared to surgical management. However, this is an association analysis unadjusted for indication bias and cannot be interpreted as a causal effect of treatment on survival. In the surgery group, survival rates were 63% at 6 months, 44.9% at one year and 36.2% at two years.

Discussion

During the years 2020 and 2021, 992 files involving 630 patients were presented to the bone metastasis MDT meeting at a university hospital. In decreasing order of frequency, the main primary cancers were bronchopulmonary (22.1%), myeloma (14.9%), breast (14.8%), prostate (9.8%) and thyroid (6.2%). Although there was a slight over-representation of hemopathies due to proximity to an expert center, these results are consistent with those of the international literature [23, 26–30]. Svensson et al. also found pulmonary cancer with bone metastasis to have the worst survival and breast cancer to have the best prognosis (they did not analyzed thyroid cancer as in this study) [26, 27]. Overall survival for the cohort from

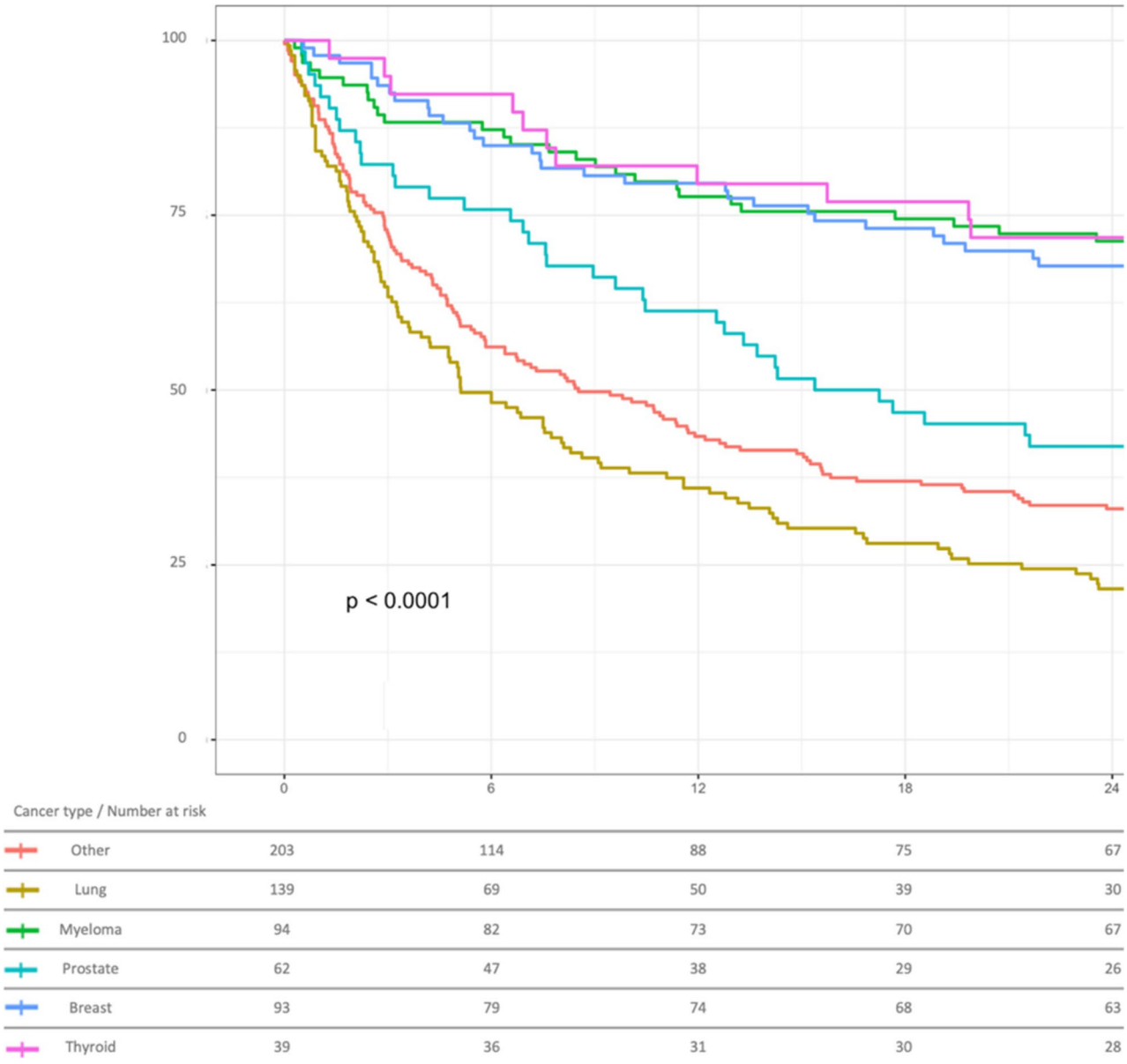


Fig. 3 Kaplan–Meier estimates of survival according to primary cancer type, expressed in % of patients alive at the 3 months endpoints

the MDT presentation was 64.1% at 6 months, 53.1% at 12 months and 43.1% at 24 months, which is slightly better than older cohort as Phanphaisarn et al. founding a median survival of 6 months for example. One of the lessons to be learned from this review is that 50% of the cohort died within 2 years of presentation to the MDT meeting.

Demographic analysis revealed a significant association between older age and poorer prognosis: the 24-month survival rate was 50.4% for patients aged ≤ 65 years compared to 39.7% for those aged > 65 years (p = 0.009). In multivariable analysis, each additional year of age was associated with a 1.6% increase in the risk of death, consistent with previous findings [31].

Gender also demonstrated a prognostic impact, as a significant survival difference between men and women was observed in both univariate and multivariable analyses, in line with existing literature [32].

Survival outcomes varied substantially by primary tumor type. Bronchopulmonary and prostate cancers were associated with the poorest survival, with median overall survival (OS) of 5.1 months and 16.3 months, respectively. In contrast, breast, thyroid, and myeloma tumors exhibited a more favorable prognosis, with median survival exceeding 24 months. This disparity may be explained by the advent of novel targeted therapies, which have markedly improved outcomes for certain disseminated cancers, particularly breast cancer [33].

However, these findings raise critical clinical considerations. Amelot et al. reported that delaying surgery for spinal metastases increases the risk of requiring intervention at a later stage due to neurological deficit [33, 34]. This highlights the need to carefully weigh the benefit-risk ratio of surgery in older patients with spinal metastases, taking into account diagnosis and life expectancy. This study found that spinal, pelvic, and other bone metastases were associated with poorer survival compared with patients without metastatic disease. Most of the literature on bone metastases does not report survival based on the location of the metastasis, and Coleman et al. did not find any significant association with localization [5]. However, spinal metastases are known to carry an increased mortality risk due to the specific complications of neurological impairment [2]. Similarly, pelvic metastases, which can limit ambulation due to pain, may also worsen the overall prognosis by leading to prolonged bed rest and immobility.

In terms of treatment, our series shows a median survival of 9.6 months after radiotherapy. There was no significant difference in survival between patients who underwent surgery and those who did not. Patients who received radiotherapy were 1.38 times more likely to die than those who did not, while those who underwent cementoplasty were 1.25 times more likely to die than those who did not. However, these findings must be interpreted with caution due to the retrospective nature of the analysis and inherent indication bias. They should not be viewed as evidence of inefficacy for cementoplasty or radiotherapy.

These results must be considered in the context of the significant frailty of the patients treated. A care plan favoring radiotherapy and cementoplasty over surgery may reflect a more palliative approach. Conversely, some surgical decisions, particularly in the spine, may be driven by the onset of neurological disorders, necessitating emergency intervention in patients with a poorer survival prognosis.

Given these findings, it is reasonable to question the timing of the treatments implemented and whether adopting more aggressive strategies at earlier disease stages might be beneficial [35]. Finally, the combination of multiple therapies—such as radiotherapy, cementoplasty, surgery, and chemotherapy—is essential for establishing an effective treatment pathway. These modalities can complement one another, potentially improving both quantitative and qualitative survival compared to their separate use [36–38]. The use of a multidisciplinary team (MDT) facilitates the coordination of different specialists to provide the optimal care pathway: nearly 80% of patients who underwent surgery also received radiotherapy during follow-up [35–38].

The purpose of MDT meetings is to discuss collegially cases, to take the best decision possible for the patient, objectively. Regarding bone metastasis, the question asked by the oncologist or referring clinician is whether this particular metastasis should be operated on, cemented, irradiated, both or neither of it. Factors to be considered are general condition with performance status, age, primary cancer, prognosis, time between diagnosis and appearance of bone lesion, associated secondary visceral lesion and associated therapies [39–41]. The mean age in this study was 65 years-old with 24% of the patients being older than 75 years-old, and due to the general aging of the population, we may expect more elderly patients with bone metastasis. Having an oncogeriatric specialist in the MDT will become mandatory, as age, if an indicator, is not necessary the true reflect of geriatric frailty. Regarding spinal metastasis, composite scores such as the Neurologic, oncologic, mechanical, and systemic (NOMS) decision framework or guidelines as ESMO clinical practices guidelines can help with management [42, 43]. PathFx is also an online algorithm clinically validated that may help clinician with decision making in bone metastasis [44]. Tokuhashi score is particularly useful to assess surgical indication in bone metastasis [45, 46]. As many tools, they might not be as used as they should be, and seeing the results of this series regarding survival, especially survival after surgery (more than one third of the cohort died within 6 months of the surgery), we are considering finding the resources to apply scoring systems systematically. The MDT seems mandatory to take decision, but clinicians should also keep an open mind to new tools that may help as AI (artificial intelligence) powered tools which may also be less time-consuming than manual entry and analysis of data [47].

This study has several limitations, particularly those inherent to its retrospective design, which may introduce data collection biases. Additionally, the analyzed files were processed during 2020 and 2021, a period marked by the COVID-19 pandemic. Although healthcare facilities strived to maintain standard care pathways, it cannot be excluded that diagnostic or therapeutic delays may have impacted patients' prognosis. This is supported by findings from Brown et al.,who reported that nearly one-third of respondents in an international survey noted delays in palliative radiotherapy during this challenging period [48]. However, the use of the INSEE database allowed us to ensure comprehensive data regarding the criterion of death or survival. Furthermore, our institution made every effort to uphold the standard of care for oncologic patients. The number of patients discussed in the MDT during these years remained comparable to 2019 and 2023.

Secondly, survival in this study was calculated from the date of the MDT meeting rather than the date of diagnosis of the secondary lesion. While this may complicate comparisons with other studies in the literature, this approach was deliberately chosen to focus on the MDT's activity and to highlight the critical turning point that the discovery of a metastatic bone lesion represents in the course of oncological disease.

As the primary endpoint of this study was survival, other factors—such as quality of life and ECOG, comorbidities, visceral metastasis, detailed oncologic history, and treatments unrelated to bone metastasis—were not analyzed. These aspects may be explored in future research but would require additional resources for data collection.

Lastly, this was a non-interventional, retrospective analysis, which introduces a significant indication bias. Indeed, for the same bone lesion, when a patient's estimated survival is less than 6 months, the MDT is less likely to opt for surgery and more likely to choose less invasive treatment options, such as radiotherapy or cementoplasty, compared to patients with a better prognosis. Therefore, the results regarding treatment efficacy must be interpreted with great caution.

The novelty of this study lies in its comprehensive, multi-tumor analysis, which addresses a significant gap in the existing literature. Unlike most published studies—typically limited to a single tumor type—our work incorporates detailed parameters, including metastasis localization and treatment modalities, which are often absent from national healthcare databases. Collectively, these elements provide a precise and representative overview of bone metastasis management practices within a large university hospital setting.

Conclusion

This study provides a comprehensive description of bone metastasis multidisciplinary team (MDT) meeting activity in a university hospital, detailing activity volume, prognostic factors, and therapeutic decisions, with a notably higher use of cementoplasty compared to historical series. To our knowledge, this represents the first French retrospective series to integrate demographic, oncological, and therapeutic data.

The median overall survival for patients with secondary bone lesions was 17 months. Lung and prostate cancers exhibited the lowest 12-month survival rates, at 36% and 61%, respectively. Poor prognostic factors identified in our cohort included male sex, age > 65 years, and the presence of metastases in the pelvis or spine.

Furthermore, close collaboration between orthopaedic surgeons and interventional radiologists is essential. This multidisciplinary approach helps limit surgical indications where feasible, particularly given the poor survival prognosis in these patients. To optimize management, clinical, histological and radiological prognostic factors must be considered when establishing prognosis and determining appropriate therapeutic strategies. Consequently, referral to a multidisciplinary consultation meeting should be mandatory.

Acknowledgements
Acknowledgments: We thank all medical practitioners contributing to the weekly MDT for their implication.

Declaration of generative AI in scientific writing
No generative AI was used to write this manuscript.

Authors' contributions
HPM, RB, LM and LMH designed the study, PC, RB, LM and LMH collected the data; PC, MK and LMH analyzed the data, PC and LMH wrote the manuscript; all authors reviewed and revised the manuscript.

Funding
None.

Data availability
The data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request. Data are located in controlled access data storage at Pitié-Salpêtrière University Hospital.

Declarations

Ethics approval and consent to participate
This study was approved and registered in the APHP (Assistance Publique Hôpitaux de Paris) registry under number 20240410163136.

Consent for publication
Consent to use anonymous data for all patients has been obtained. The study has been performed according to the Declaration of Helsinki.

Competing interests
The authors declare no competing interests.

Author details
[1]Orthopaedic Surgery Unit, Pitié-Salpêtrière University Hospital, Assistance Publique - Hôpitaux de Paris, 47, Boulevard de l'hôpital, Paris 75013, France
[2]Faculté de Médecine, Sorbonne Université, Paris, France
[3]Orthopaedic Surgery Unit, Hôpital Européen Georges Pompidou, Assistance Publique - Hôpitaux de Paris, 20, rue Leblanc, Paris 75015, France
[4]Université Paris-Cité, Paris, France
[5]Department of Medical Oncology, Pitié-Salpêtrière University Hospital, Assistance Publique - Hôpitaux de Paris, 47, Boulevard de l'hôpital, Paris 75013, France

Received: 20 October 2025 / Accepted: 29 July 2026
Published online: 01 August 2026

References

1. Le réseau FRANCIM. Les registres des cancers en France. Nantes, Santé Pays de la Loire; 2015.
2. Sailhan F, Prost S, Zairi F, Gille O, Pascal-Mousselard H, Bennis S, et al. Retrospective multicenter study by the French Spine Society of surgical treatment for spinal metastasis in France. Orthop Traumatology: Surg Res. 2018;104:589–95. https://doi.org/10.1016/j.otsr.2018.06.006.
3. Choi D, Bilsky M, Fehlings M, Fisher C, Gokaslan Z. Spine Oncology—Metastatic Spine Tumors Neurosurg. 2017;80:S131–7. https://doi.org/10.1093/neuros/nyw084.

4. Guo M, Kolberg KL, Smith EC, Smith BW, Yousif JE, Kessler JL, et al. Predominance of Spinal Metastases Involving the Posterior Vertebral Body. World Neurosurg. 2018;119:e991–6. https://doi.org/10.1016/j.wneu.2018.08.029.
5. Coleman RE. Metastatic bone disease: clinical features, pathophysiology and treatment strategies. Cancer Treat Rev. 2001;27:165–76. https://doi.org/10.1053/ctrv.2000.0210.
6. Mercadante S. Malignant bone pain: pathophysiology and treatment. Pain. 1997;69:1–18. https://doi.org/10.1016/s0304-3959(96)03267-8.
7. DiCaprio MR, Murtaza H, Palmer B, Evangelist M. Narrative review of the epidemiology, economic burden, and societal impact of metastatic bone disease. Ann Jt. 2022;7:28–28. https://doi.org/10.21037/aoj-20-97.
8. Glare P. A systematic review of physicians' survival predictions in terminally ill cancer patients. BMJ. 2003;327:195–0. https://doi.org/10.1136/bmj.327.7408.195.
9. Christakis NA. Extent and determinants of error in physicians' prognoses in terminally ill patients: prospective cohort study. West J Med. 2000;172:310–3. https://doi.org/10.1136/ewjm.172.5.310.
10. Hartsell WF, Desilvio M, Bruner DW, Scarantino C, Ivker R, Roach M, et al. Can Physicians Accurately Predict Survival Time in Patients with Metastatic Cancer? Analysis of RTOG 97 – 14. J Palliat Med. 2008;11:723–8. https://doi.org/10.1089/jpm.2007.0259.
11. Choi D, Ricciardi F, Arts M, Buchowski JM, Bunger C, Chung CK, et al. Prediction Accuracy of Common Prognostic Scoring Systems for Metastatic Spine Disease. Spine (Phila Pa 1976). 2018;43:1678–84. https://doi.org/10.1097/BRS.0000000000002576.
12. Chow E, Zeng L, Salvo N, Dennis K, Tsao M, Lutz S. Update on the Systematic Review of Palliative Radiotherapy Trials for Bone Metastases. Clin Oncol. 2012;24:112–24. https://doi.org/10.1016/j.clon.2011.11.004.
13. McDonald R, Ding K, Brundage M, Meyer RM, Nabid A, Chabot P, et al. Effect of Radiotherapy on Painful Bone Metastases. JAMA Oncol. 2017;3:953. https://doi.org/10.1001/jamaoncol.2016.6770.
14. van der Linden YM, Lok JJ, Steenland E, Martijn H, van Houwelingen H, Marijnen CAM, et al. Single fraction radiotherapy is efficacious: a further analysis of the Dutch Bone Metastasis Study controlling for the influence of retreatment. Int J Radiation Oncology*Biology*Physics. 2004;59:528–37. https://doi.org/10.1016/j.ijrobp.2003.10.006.
15. Dennis K, Wong K, Zhang L, Culleton S, Nguyen J, Holden L, et al. Palliative Radiotherapy for Bone Metastases in the Last 3 Months of Life: Worthwhile or Futile? Clin Oncol. 2011;23:709–15. https://doi.org/10.1016/j.clon.2011.05.004.
16. Chow E, Fung K, Panzarella T, Bezjak A, Danjoux C, Tannock I. A predictive model for survival in metastatic cancer patients attending an outpatient palliative radiotherapy clinic. Int J Radiation Oncology*Biology*Physics. 2002;53:1291–302. https://doi.org/10.1016/S0360-3016(02)02832-8.
17. Steinvoort-Draat IN, Otto-Vollaard L, Quint S, Tims JL, de Pree IMN, Nuyttens JJ. Palliative radiotherapy: New prognostic factors for patients with bone metastasis. Cancer Radiother. 2024;28:236–41. https://doi.org/10.1016/j.canrad.2023.09.003.
18. Wu SY, Singer L, Boreta L, Garcia MA, Fogh SE, Braunstein SE. Palliative radiotherapy near the end of life. BMC Palliat Care. 2019;18:29. https://doi.org/10.1186/s12904-019-0415-8.
19. Westhoff PG, de Graeff A, Monninkhof EM, Bollen L, Dijkstra SP, van der Steen-Banasik EM, et al. An Easy Tool to Predict Survival in Patients Receiving Radiation Therapy for Painful Bone Metastases. Int J Radiation Oncology*Biology*Physics. 2014;90:739–47. https://doi.org/10.1016/j.ijrobp.2014.07.051.
20. Prost S, Bouthors C, Fuentes S, Charles Y-P, Court C, Mazel C, et al. Influence of preoperative biological parameters on postoperative complications and survival in spinal bone metastasis. A multicenter prospective study. Orthop Traumatol Surg Res. 2020;106:1033–8. https://doi.org/10.1016/j.otsr.2019.11.031.
21. Stacoffe N, Mesny E, Proriol M, Fontana A, Chalamet B, BenRejeb M-I, et al. Interventional radiology in bone metastases: current concepts and perspectives. Joint Bone Spine. 2026;106076. https://doi.org/10.1016/j.jbspin.2026.106076.
22. Vieillard M-H, Thureau S. Réunions de concertation pluridisciplinaire (RCP) dédiées aux métastases osseuses: historique et justification. Bull Cancer. 2013;100:1135–9. https://doi.org/10.1684/bdc.2013.1851.
23. Hong S, Youk T, Lee SJ, Kim KM, Vajdic CM. Bone metastasis and skeletal-related events in patients with solid cancer: A Korean nationwide health insurance database study. PLoS ONE. 2020;15:e0234927. https://doi.org/10.1371/journal.pone.0234927.
24. Bădilă AE, Rădulescu DM, Niculescu A-G, Grumezescu AM, Rădulescu M, Rădulescu AR. Recent Advances in the Treatment of Bone Metastases and Primary Bone Tumors: An Up-to-Date Review. Cancers (Basel). 2021;13:4229. https://doi.org/10.3390/cancers13164229.
25. Kaplan EL, Meier P. Nonparametric Estimation from Incomplete Observations. J Am Stat Assoc. 1958;53:457–81. https://doi.org/10.1080/01621459.1958.10501452.
26. Phanphaisarn A, Patumanond J, Settakorn J, Chaiyawat P, Klangjorhor J, Pruksakorn D. Prevalence and Survival Patterns of Patients with Bone Metastasis from Common Cancers in Thailand. Asian Pac J Cancer Prev. 2016;17:4335–40.
27. Svensson E, Christiansen CF, Ulrichsen SP, Rørth MR, Sørensen HT. Survival after bone metastasis by primary cancer type: a Danish population-based cohort study. BMJ Open. 2017;7:e016022. https://doi.org/10.1136/bmjopen-2017-016022.
28. Hernandez RK, Wade SW, Reich A, Pirolli M, Liede A, Lyman GH. Incidence of bone metastases in patients with solid tumors: analysis of oncology electronic medical records in the United States. BMC Cancer. 2018;18:44. https://doi.org/10.1186/s12885-017-3922-0.
29. Singh V, Haseeb A, Alkubaisi A. Incidence and outcome of bone metastatic disease at University Malaya Medical Centre. Singap Med J. 2014;55:539–46. https://doi.org/10.11622/smedj.2014138.
30. Maccauro G, Spinelli MS, Mauro S, Perisano C, Graci C, Rosa MA. Physiopathology of Spine Metastasis. Int J Surg Oncol. 2011;2011:1–8. https://doi.org/10.1155/2011/107969.
31. Lo W-Y, Yang S-H. Metastatic spinal cord compression (MSCC) treated with palliative decompression: Surgical timing and survival rate. PLoS ONE. 2017;12:e0190342. https://doi.org/10.1371/journal.pone.0190342.
32. Ryan C, Stoltzfus KC, Horn S, Chen H, Louie AV, Lehrer EJ, et al. Epidemiology of bone metastases. Bone. 2022;158:115783. https://doi.org/10.1016/j.bone.2020.115783.
33. Corona SP, Sobhani N, Ianza A, Roviello G, Mustacchi G, Bortul M, et al. Advances in systemic therapy for metastatic breast cancer: future perspectives. Med Oncol. 2017;34:119. https://doi.org/10.1007/s12032-017-0975-5.
34. Amelot A, Balabaud L, Choi D, Fox Z, Crockard HA, Albert T, et al. Surgery for metastatic spine tumors in the elderly. Advanced age is not a contraindication to surgery! Spine J. 2017;17:759–67. https://doi.org/10.1016/j.spinee.2015.07.440.
35. Tabourel G, Terrier L-M, Dubory A, Cristini J, Nail L-R, Le, Cook A-R, et al. Are spine metastasis survival scoring systems outdated and do they underestimate life expectancy? Caution in surgical recommendation guidance. J Neurosurg Spine. 2021;35:527–34. https://doi.org/10.3171/2020.12.SPINE201741.
36. Uei H, Tokuhashi Y, Maseda M, Nakahashi M, Sawada H, Nakayama E, et al. Clinical results of multidisciplinary therapy including palliative posterior spinal stabilization surgery and postoperative adjuvant therapy for metastatic spinal tumor. J Orthop Surg Res. 2018;13:30. https://doi.org/10.1186/s13018-018-0735-z.
37. Lee C-H, Kwon J-W, Lee J, Hyun S-J, Kim K-J, Jahng T-A, et al. Direct Decompressive Surgery Followed by Radiotherapy Versus Radiotherapy Alone for Metastatic Epidural Spinal Cord Compression. Spine (Phila Pa 1976). 2014;39:E587–92. https://doi.org/10.1097/BRS.0000000000000258.
38. Hirabayashi H, Ebara S, Kinoshita T, Yuzawa Y, Nakamura I, Takahashi J, et al. Clinical outcome and survival after palliative surgery for spinal metastases. Cancer. 2003;97:476–84. https://doi.org/10.1002/cncr.11039.
39. Choi D, Fox Z, Albert T, Arts M, Balabaud L, Bunger C, et al. Prediction of Quality of Life and Survival After Surgery for Symptomatic Spinal Metastases. Neurosurgery. 2015;77:698–708. https://doi.org/10.1227/NEU.0000000000000907.
40. Verlaan J-J, Choi D, Versteeg A, Albert T, Arts M, Balabaud L, et al. Characteristics of Patients Who Survived < 3 Months or > 2 Years After Surgery for Spinal Metastases:Can We Avoid Inappropriate Patient Selection? J Clin Oncol. 2016;34:3054–61. https://doi.org/10.1200/JCO.2015.65.1497.
41. Lei M, Li J, Liu Y, Jiang W, Liu S, Zhou S. Who are the Best Candidates for Decompressive Surgery and Spine Stabilization in Patients With Metastatic Spinal Cord Compression? Spine (Phila Pa 1976). 2016;41:1469–76. https://doi.org/10.1097/BRS.0000000000001538.
42. Newman WC, Laufer I, Bilsky MH, Neurologic. Oncologic, Mechanical, and Systemic and Other Decision Frameworks for Spinal Disease. Neurosurg Clin N Am. 2020;31:151–66. https://doi.org/10.1016/j.nec.2019.11.005.
43. Coleman R, Body JJ, Aapro M, Hadji P, Herrstedt J, ESMO Guidelines Working Group. Bone health in cancer patients: ESMO Clinical Practice Guidelines. Ann Oncol. 2014;25(Suppl 3):iii124–37. https://doi.org/10.1093/annonc/mdu103.
44. Anderson AB, Wedin R, Fabbri N, Boland P, Healey J, Forsberg JA. External Validation of PATHFx Version 3.0 in Patients Treated Surgically and Nonsurgically for Symptomatic Skeletal Metastases. Clin Orthop Relat Res. 2020;478:808–18. https://doi.org/10.1097/CORR.0000000000001081.

45. Gakhar H, Swamy GN, Bommireddy R, Calthorpe D, Klezl Z. A study investigating the validity of modified Tokuhashi score to decide surgical intervention in patients with metastatic spinal cancer. Eur Spine J. 2013;22:565–8. https://doi.org/10.1007/s00586-012-2480-z.
46. Shimada H, Setoguchi T, Nakamura S, Yokouchi M, Ishidou Y, Tominaga H, et al. Evaluation of prognostic scoring systems for bone metastases using single-center data. Mol Clin Oncol. 2015;3:1361–70. https://doi.org/10.3892/mco.2015.637.
47. Papalia GF, Brigato P, Sisca L, Maltese G, Faiella E, Santucci D, et al. Artificial Intelligence in Detection, Management, and Prognosis of Bone Metastasis: A Systematic Review. Cancers (Basel). 2024;16. https://doi.org/10.3390/cancers16152700.
48. Brown JE, Wood SL, Confavreux C, Abe M, Weilbaecher K, Hadji P, et al. Management of bone metastasis and cancer treatment-induced bone loss during the COVID-19 pandemic: An international perspective and recommendations. J Bone Oncol. 2021;29:100375. https://doi.org/10.1016/j.jbo.2021.100375.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.